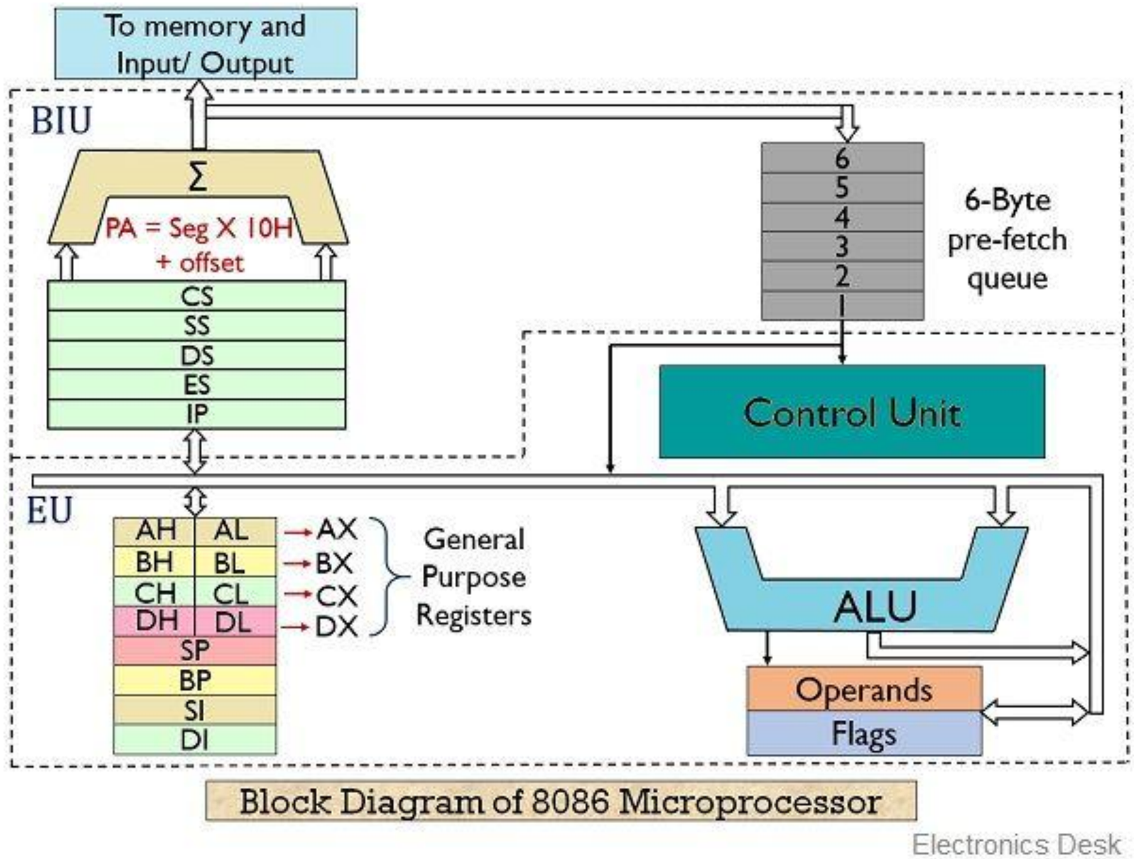
8086 Microprocessor Block Diagram – Step-by-Step Explanation



Block Diagram of 8086 Microprocessor

Basic Details

Processor Type: 16-bit processor (processes 16 bits of data at a time)

Address Bus: 20 bits – can access $2^{20}=1,048,576$220=1,048,576 memory locations

Memory Supported: **1MB (1 megabyte)**

Package: **40-pin** Dual Inline Package (DIP)

Power Supply: 5V DC

Clock Speed:

8086: 5 MHz

8086-2: 8 MHz

8086-1: 10 MHz

Main Blocks in the 8086 Microprocessor

1. Bus Interface Unit (BIU)

- Handles all communication with memory and input/output devices
- Fetches instructions from memory and reads/writes data
- Computes physical memory addresses using segment and offset registers
- Components:
- Segment Registers: Code, Data, Stack, and Extra Segment
- Instruction Queue: Prefetches up to 6 bytes of instructions (pipelining)
- Address Generation Circuitry: Forms 20-bit physical address

2. Execution Unit (EU)

- Executes instructions fetched by the BIU
- Components:
- Arithmetic Logic Unit (ALU): Performs arithmetic and logic operations
- General Purpose Registers: AX, BX, CX, DX
- Flag Register: Shows the result of operations (zero, carry, sign, etc.)
- Instruction Decoder: Decodes instructions for the ALU
- Pointer and Index Registers: SP, BP, SI, DI (used for addressing in memory operations)

3. Communication between BIU and EU

- While the BIU fetches the next instruction (stores in the queue), the EU executes the current instruction
- This is known as a **pipelined architecture**, which allows both units to work in parallel for faster operation

Additional Features

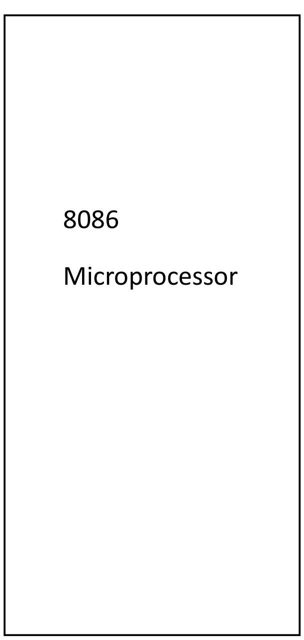
Multiplexed Address/Data Bus: Lower 16 pins (AD0–AD15) are used for both address and data (depends on clock cycle)

Interrupt System: Handles external and internal interrupts, allowing urgent tasks to be managed efficiently

Clock Generator: Needs an external clock signal to time and synchronize all operations

Summary Table

Feature	Description
Data Bus Width	16 bits
Address Bus Width	20 bits
Max Memory Supported	1MB
Package	40-pin DIP
Power Supply	5V
Clock Speed	5, 8, or 10 MHz
Main Blocks	Bus Interface Unit (BIU), Execution Unit (EU)
Internal Registers	AX, BX, CX, DX, SP, BP, SI, DI, (Code/Data/Stack/Extra Segment-**Starting Address**)
Instruction Queue	6 bytes (prefetch for pipelining)
Bus Multiplexing	Address/Data lines are shared (AD0–AD15)



Memory Segmentation